PRODUCTION RATE PREDICTION IN ASSEMBLY LINE

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GOAL:

To predict production output based on line parameters, you collect historical data that includes production results and relevant input variables like machine speed, number of workers, shift duration, and downtime. This data is cleaned and prepared, with numerical values standardized and categorical variables encoded. A regression model—such as linear regression, random forest, or gradient boosting—is then trained to learn the relationship between inputs and output. After evaluating the model's accuracy using metrics like MAE or R^2, it can be deployed to predict future production outputs based on current line settings, with regular monitoring to ensure performance over time.

DATA SET:

Shift length, No. of operators, Machine availability, Operator efficiency, Downtime, Task complexity, Material supply delay, Line balancing index, Rework rate, Automation level.

https://in.docworkspace.com/d/sINil9NeWAuqnxsAG?sa=601.1074

ALGORITHM USED:

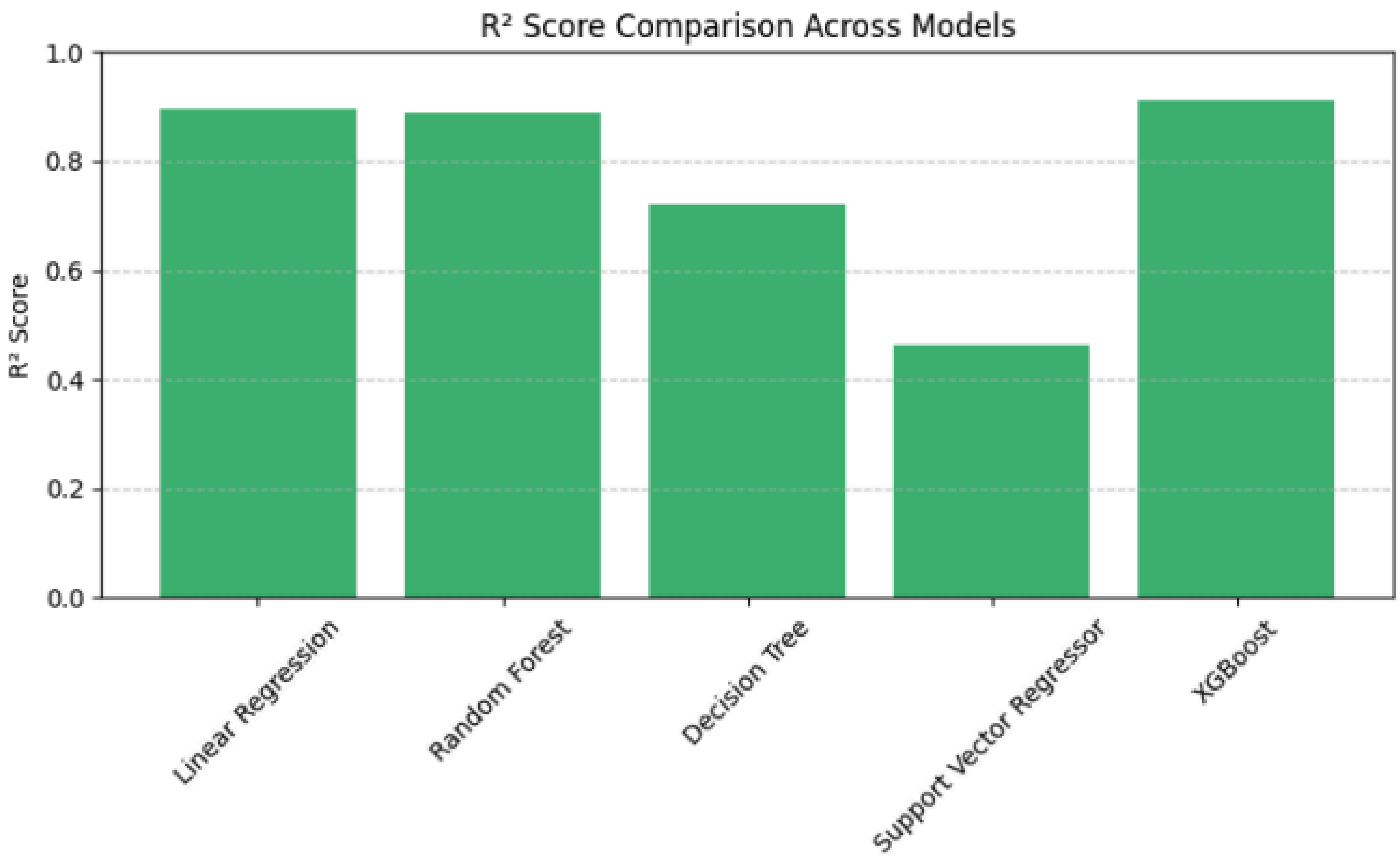
Linear regression

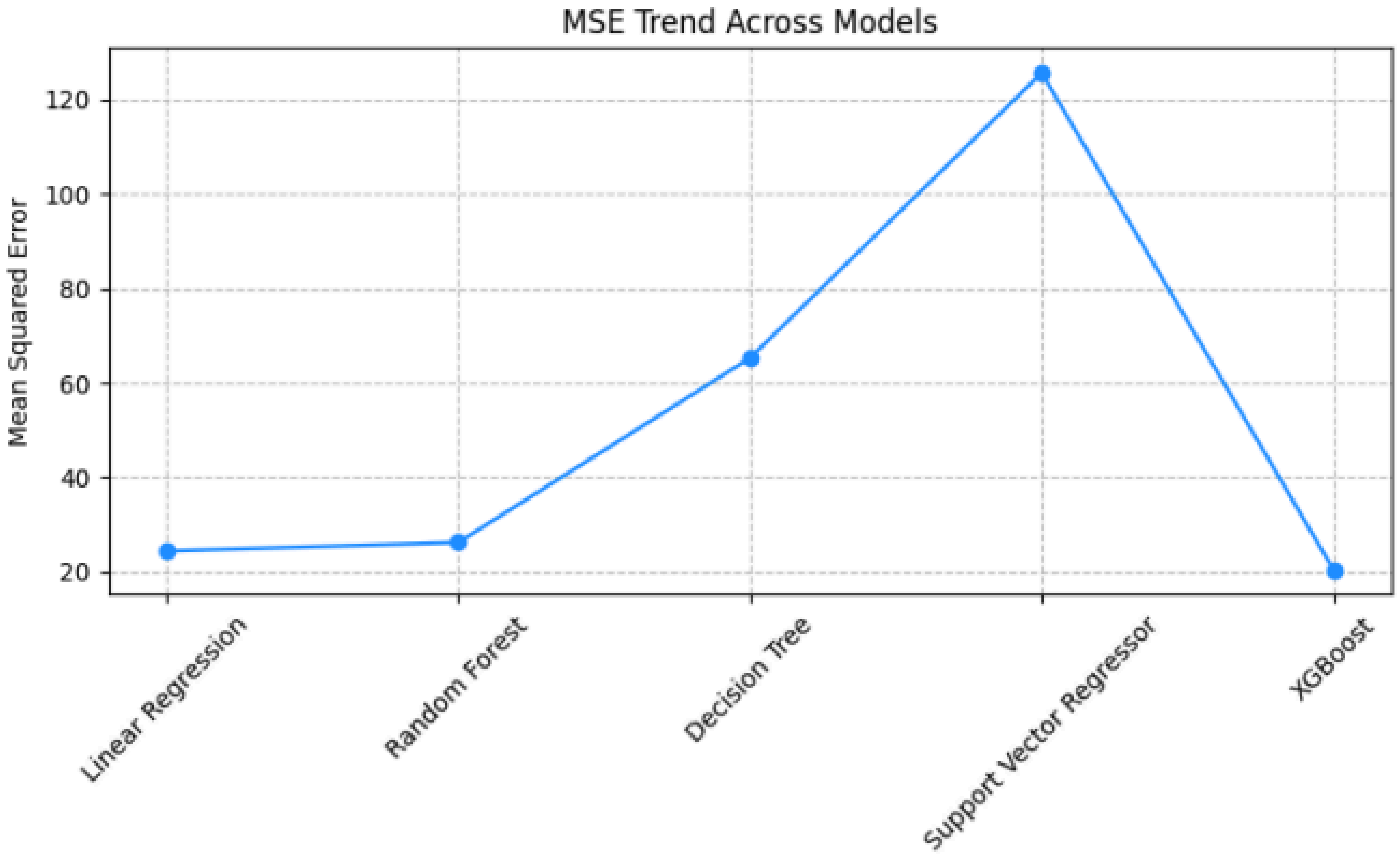
Random forest

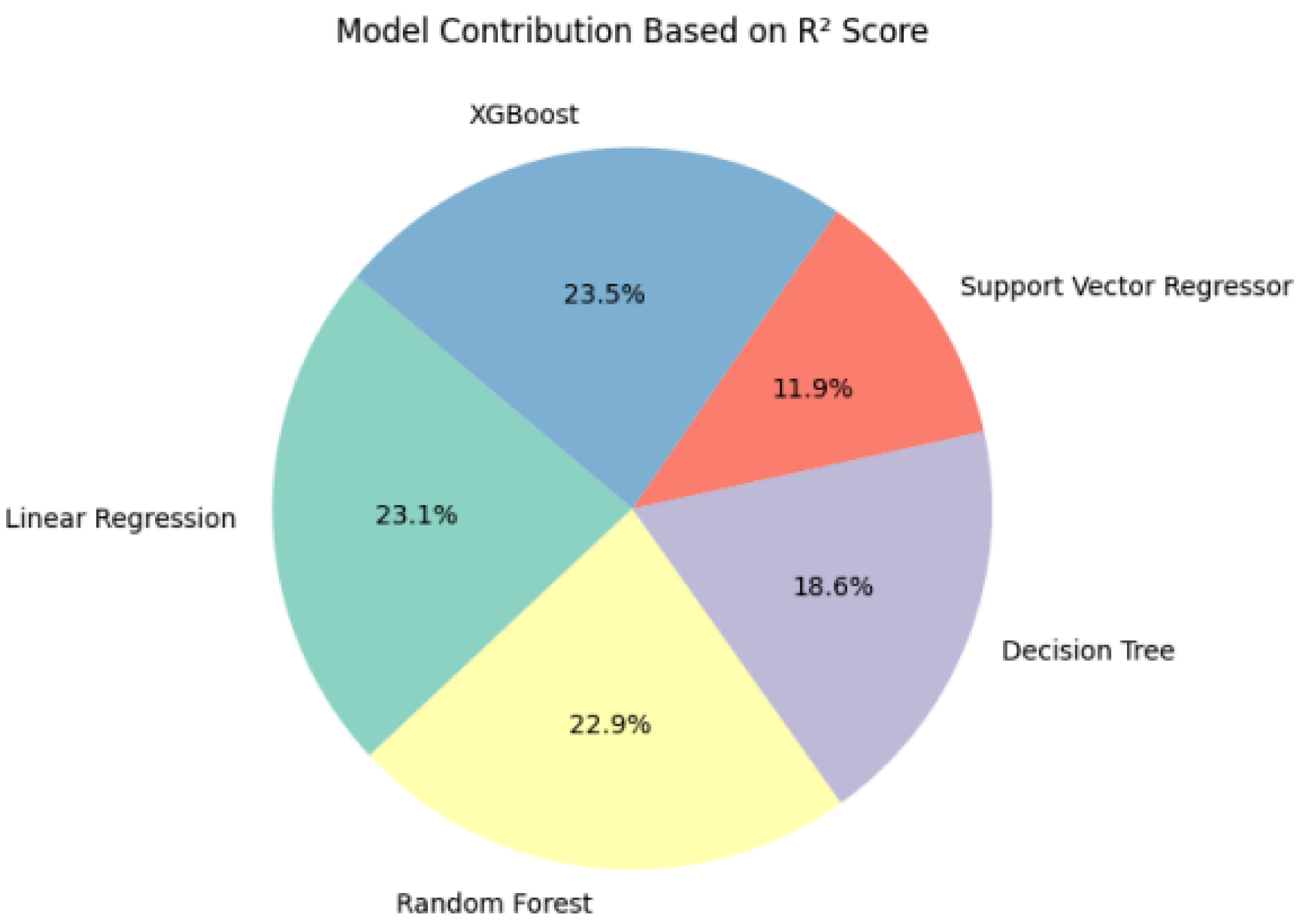
Decision tree

Support vector regression

XGBoost


R² Score Comparison Across Models
R² Score
1.0
0.8
0.6
0.4
0.2
0.0
Linear Regression
Random Forest
Decision Tree
Support Vector Regressor
XGBoost


MSE Trend Across Models
Mean Squared Error
20
40
60
80
100
120
Linear Regression
Random Forest
Decision Tree
Support Vector Regressor
XGBoost


Model Contribution Based on R² Score
XGBoost
23.5%
Support Vector Regressor
11.9%
Linear Regression
23.1%
18.6%
Decision Tree
22.9%
Random Forest

Why XGBoost Is Best ?

XGBoost is often considered one of the best algorithms for predicting production rates in assembly lines due to its high accuracy, speed, and ability to handle complex, non-linear relationships between input parameters and output. It can manage a mix of numerical and categorical data, handle missing values gracefully, and automatically capture interactions between features without extensive manual feature engineering. Additionally, XGBoost uses ensemble learning and regularization techniques to reduce overfitting, making it robust for real-world industrial data, which is often noisy and unbalanced. Its scalability and efficiency also make it suitable for large datasets typical in manufacturing environments.

BENEFITS:

Predicting production output based on line parameters offers several real-life benefits in manufacturing and operations management. It enables proactive decision-making by allowing managers to adjust inputs like staffing or shift lengths to meet production targets efficiently. This predictive insight helps reduce downtime, optimize resource allocation, and improve scheduling accuracy. It also supports identifying bottlenecks and areas for improvement, such as underperforming operators or inefficiencies due to task complexity or poor line balancing. Ultimately, it leads to increased productivity, reduced operational costs, improved product delivery timelines, and more data-driven strategic planning.

Thank You